E8: Every Combination of Five Head Filters

Lexical task heads, simple tasks

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In short. E8 describes each head by five yes/no properties: it decodes the task's most active words under J-head (D), it does not look mainly at BOS (B), it attends to the instruction's task words (T), to the query (I), or not mainly to the rest of the instruction (R). Every combination was patched under E4's held-out protocol. Two sets stand out. Attention to the task words alone (T) is the best E8 set and the one chosen on the first fold. Decoding plus the BOS filter (DB) comes next with less than half as many heads; it needs no hand-written word list and no causal step. It is above CC-pooled and FV-20 and below FV-N_t. The two win different tasks: T the facts, DB the word relations, present–past and French. Attention to the query or to the rest of the instruction lowers every set it is added to. A first run had missed the country decoders: our lowercased word lists never matched the " Capital" tokens the lens decodes. With that fixed and the list lengthened to 30 words, every set that uses D improved, DB the most.

1 Why

E7 showed that among heads that decode a task's words, the ones that look at the prompt rather than BOS carry the task. That leaves open which attention property matters: is "not BOS" the right filter, or do heads that attend to the task words, or to the item the task is applied to, do better? And does decoding add anything once attention is known? E8 tests every combination of one decoding filter and four attention filters, on the same items as E4 and E7.

2 Design

E4's protocol, J-head only, rule style, the 14 simple tasks. All filters and thresholds were fixed before patching; the list behind D was changed after the first run (Sec. 3).

1. **Filters**, per task and head, on the choosing half's rule prompts (five wordings pooled, 500 prompts), at the last prompt token:
   - **D** decoding: share of prompts whose J-head top 10 holds a token of the task's word list; pass if $\geq$0.10 (the CC / E7 head rule). List: the task's 30 most active decoded words, E7's word rules (top-10 tokens decoded, lowercased, alphabetic, $\geq$3 letters, no stop words, Zipf $\geq$3; activity = share of (prompt, head) pairs whose top 10 holds the word), nothing removed.
   - **B** BOS: mean attention on BOS $\leq$0.5 (E7's filter).
   - **T** task words: mean attention on the instruction's task-word tokens $\geq$0.05. Task words = Yang et al.'s term list plus the task's top-10 J-head "specific" words of E4 (choosing half), so T uses a hand-written list and the lens.
   - **I** input: mean attention on the query $\geq$0.05.
   - **R** rest: mean attention on the rest of the instruction $\leq$0.5.
2. **Sets**: a set is named by its filters (DB = D and B); a head is in it if it passes all of them. All 31 non-empty combinations, all layers, no size cap; an empty set is left out for that task. No random controls.
3. **Patching** (E4): both folds; rule means of the choosing half and the control mean in both checks; $\alpha \in \{1, 2\}$ only (to save compute; E7's chosen α were mostly 1–2); replies $\leq$16 tokens; flexible score; every answer saved per item. Check 1: items answered correctly with one shot, patched in their zero-shot prompts. Check 2: items answered wrongly with one shot, patched in their 1-shot prompts. Accuracy at the cross-fitted α (the best one on the other fold), mean of the two folds.
4. **References** on the same items and folds: E7's raw +F, E4's CC-20, CC, FV-20 and FV-N_t. Their runs had ten α; here they are cross-fitted over α 1 and 2 only, so they are a little lower than in their own reports (CC 0.40 vs 0.44, FV-20 0.33 vs 0.38, raw +F 0.31 vs 0.40 in check 1).

5. **Filter choice without the reported items**: the set with the highest check-1 accuracy on fold 1 (each task at its own best α) is picked; its fold-2 accuracy is the reported number.

3 The decoding filter: a matching gap, and the list length

First run. D first used E7's raw list (the 10 most active words) and the word-to-token rule of Yang et al.'s code: each word is tried as written, with a leading space, in upper, capitalised and lower case, and after ".", "_" or "-". D sets collapsed on landmark–country (0.02–0.03 in check 1). Looking at the heads behind it: L21H2 has " England" in its J-head top 10 on 488 of 500 prompts and *england* is on the list, yet its D rate was 0. The rule never forms a leading space with a capital letter, which is how the lens decodes names. Yang et al.'s own lists are capitalised, so their code does not meet the problem; our data-driven lists are lowercased.

Fix. For our lowercased lists, the leading-space capitalised and upper-case forms are added (" Canada", " CANADA"; `lth.terms.expand_lowercase_to_token_ids`). The Yang reproduction keeps the exact rule.

List length. Ten words is short: on landmark–country the list holds six country names, so heads decoding " Japan" or " Italy" are missed for a second reason. Before any rerun, list lengths of 10, 30 and 50 words were compared on the stored readings of the choosing halves, without patching, by how many of E4's FV-20 heads (chosen on the same half) end up in D and DB (Table 1).

words	case fix	D	FV-20 in D	chance	DB	FV-20 in DB	chance	generic
10	no	112	5.2	2.2	32	4.7	0.6	4/10
10	yes	137	6.9	2.7	41	6.0	0.8	4/10
30	no	213	8.1	4.2	56	7.1	1.1	13/30
30	yes	245	10.0	4.8	65	8.5	1.3	13/30
50	no	273	9.3	5.3	70	7.8	1.4	23/50
50	yes	310	11.2	6.1	80	9.3	1.6	23/50

Table 1: D-filter variants, mean over tasks and folds. Chance: FV-20 heads expected in a random set of the same size. Generic: list words among the 50 words heads decode most on WikiText. Source: `scripts/e8_list_length.py`.

The case fix adds FV heads at every length without lowering the ratio to chance. A longer list adds more, but less selectively: at 50 words almost half the list is generic and D holds 30% of all heads; from 30 to 50 words DB grows by 15 heads for about one more FV head. We chose 30 words with the fix. This choice was prompted by a patched result (the collapse), but made without patched accuracies. The 15 sets without D were unchanged; the 16 with D were rebuilt and rerun on the same items.

set	heads, 1st	heads, now	check 1, 1st	check 1, now	check 2, 1st	check 2, now
DB	32	65	0.31	0.51	0.34	0.32
DT	34	64	0.27	0.49	0.32	0.35
DBT	20	38	0.23	0.49	0.32	0.35
D	110	249	0.34	0.41	0.32	0.26
DTR	30	59	0.18	0.27	0.28	0.30
DBTR	16	30	0.17	0.25	0.27	0.31
DBI	21	43	0.09	0.25	0.30	0.24
DBR	23	46	0.13	0.22	0.28	0.28
DI	48	107	0.12	0.22	0.29	0.26
DTI	20	38	0.08	0.21	0.28	0.29
DR	103	229	0.16	0.20	0.28	0.25
DBTI	13	23	0.06	0.18	0.27	0.30
DTIR	18	34	0.06	0.13	0.25	0.27
DIR	44	98	0.06	0.09	0.26	0.23
DBTIR	11	20	0.05	0.09	0.23	0.27
DBIR	16	35	0.03	0.08	0.23	0.22

Table 2: Every set with D, first run (10 words, Yang's expansion) and now (30 words, with the fix): heads (median over tasks), task accuracy at the cross-fitted α, mean over tasks.

All 16 sets rise in check 1, those with B or T the most; sets roughly double in size. In check 2 the changes are small and go both ways.

4 Word lists and sets

Task	30 most active decoded words (fold 1, most active first)	fold 2 same
antonym	*opposite, what, which, word, instead, etc, because, http, opposing, words, therefore, amongst, definition, colour, reverse, synonyms, https, why, mph, maybe, only, whats, name, their, itself, they, meaning, american, ant, answer*	28
synonym	*what, which, etc, synonyms, amongst, https, word, maybe, whilst, their, words, colour, similar, synonym, http, definition, name, whats, english, then, syn, they, why, same, mph, afterward, possibly, herself, her, being*	28
singular–plural	*because, what, single, which, https, http, name, noun, etc, multiple, plural, mph, maybe, amongst, either, linguistic, therefore, names, named, then, naming, whilst, even, only, colour, similar, english, called, there, whats*	30
present–past	*what, etc, english, https, verb, http, which, grammar, name, maybe, because, linguistic, really, question, verbs, they, why, colour, mph, even, then, amongst, their, called, word, how, only, britain, whats, british*	29
country–capital	*city, cities, which, country, what, town, countries, answer, mph, international, only, maybe, http, https, geography, downtown, colour, capital, canada, really, american, there, itself, they, pretty, answers, italy, london, etc, our*	28
country–currency	*currency, currencies, country, which, monetary, money, what, countries, dollar, international, http, https, financial, maybe, american, answer, amongst, foreign, colour, whilst, mph, they, germany, really, etc, british, even, canada, only, also*	28
landmark–country	*country, which, countries, england, canada, germany, what, france, location, australia, city, international, geography, local, italy, located, mph, http, maybe, geographical, locations, etc, https, usa, county, geographic, being, cities, netherlands, amongst*	29
park–country	*country, which, countries, what, england, canada, location, australia, park, located, france, national, mph, tourism, germany, geography, unesco, parks, international, italy, city, nation, etc, tourist, geographical, county, itself, also, wildlife, locations*	30
product–company	*company, companies, which, manufacturer, what, product, http, etc, maybe, microsoft, mph, https, manufacturers, products, because, only, ones, really, who, itself, named, name, even, why, colour, owned, wikipedia, software, apple, ownership*	29
next-item	*next, which, what, january, july, september, november, october, february, mph, maybe, sequence, because, etc, question, items, december, june, after, item, april, herself, pretty, even, calendar, then, answer, amongst, christmas, really*	24
prev-item	*previous, what, preceding, prior, answer, question, item, which, next, because, mph, why, etc, items, amongst, sequence, january, previously, http, even, earlier, september, maybe, answers, https, how, november, october, she, whilst*	25
English–French	*french, france, english, paris, what, which, translation, http, etc, their, italian, european, american, they, translations, language, maybe, https, linguistic, then, japanese, german, being, colour, called, bilingual, therefore, languages, how, named*	29
English–German	*german, germany, germans, english, which, http, what, berlin, european, french, etc, american, translation, linguistic, language, https, maybe, their, called, colour, therefore, italian, von, japanese, munich, czech, translations, languages, being, die*	30
English–Spanish	*spanish, english, what, mexican, http, hispanic, translation, bilingual, spain, which, language, etc, mexico, their, translations, https, portuguese, linguistic, french, called, languages, american, being, name, they, maybe, madrid, santiago, why, therefore*	28

Table 3: D's word lists, fold 1; last column: words also on fold 2's list.

The folds agree on 24–30 of the 30 words. Between 6 and 18 words per list are generic (*what, which, http, mph, maybe*); the rest name the task or its answers (country names on landmark and park, months on next-item, *opposite* on antonym).

Task	D	B	T	I	R	DB	DT	BT	DBT	DBTIR	D (1st)	DB (1st)
antonym	241	192	206	360	970	67	69	110	44	26	124	42
synonym	246	167	156	322	971	56	58	93	38	22	134	34
singular–plural	230	210	124	332	948	67	48	79	32	19	119	38
present–past	221	200	13	337	916	52	3	12	3	0	108	28
country–capital	269	174	200	320	968	64	84	95	44	28	117	32
country–currency	262	162	149	254	960	52	61	74	32	18	106	27
landmark–country	264	212	85	364	944	84	43	47	24	15	105	32
park–country	210	210	113	451	958	68	45	58	24	16	116	36
product–company	274	199	57	442	958	82	30	40	21	14	132	35
next-item	264	160	302	359	990	64	99	104	47	38	106	25
prev-item	252	224	150	288	924	74	66	98	38	28	106	36
English–French	254	193	212	326	952	66	74	107	41	22	112	28
English–German	224	194	218	310	948	58	74	112	40	21	80	18
English–Spanish	222	190	230	286	957	62	77	114	44	19	95	30
median	249	194	154	329	958	65	64	94	38	20	110	32

Table 4: Heads per set, mean of the two folds (1,024 heads in all). D (1st), DB (1st): the first run's D and DB.

R passes nearly every head and I about a third, so neither selects much on its own. T varies most across tasks: 13 heads on present–past, about 300 on next-item. DB keeps a quarter to a third of D on every task (52–84 heads).

5 Results

5.1 Check 1

Task	items	unp.	D	B	T	DB	DT	BT	DBT	E7 raw+F	CC-20	CC	FV-20	FV-N_t
antonym	100/100	0.00	**0.74**	0.00	0.60	0.70	0.65	0.34	0.61	0.46	0.00	0.42	0.02	0.49
synonym	100/100	0.03	0.49	0.01	0.18	0.38	0.32	0.14	0.27	0.30	0.09	0.34	0.18	**0.55**
singular–plural	91/88	0.00	0.80	0.02	0.80	0.78	0.70	0.82	0.80	**0.88**	0.09	0.69	0.44	0.85
present–past	100/100	0.00	0.44	0.02	0.00	0.54	0.00	0.00	0.00	0.17	0.00	0.04	0.64	**0.85**
country–capital	84/86	0.00	0.65	0.46	**0.98**	0.69	0.89	0.93	0.94	0.87	0.77	0.75	0.88	0.94
country–currency	67/65	0.00	0.28	0.40	**0.61**	0.45	0.30	0.58	0.27	0.32	0.14	0.29	0.00	0.47
landmark–country	100/100	0.01	0.23	0.77	**0.93**	0.73	0.89	0.88	0.81	0.03	0.53	0.92	0.92	0.93
park–country	100/100	0.00	0.47	0.17	0.67	0.47	0.54	0.63	0.55	0.42	0.40	0.73	0.82	**0.88**
product–company	100/100	0.07	0.21	0.51	0.82	0.40	0.76	0.83	0.74	0.37	0.27	0.24	0.64	**0.91**
next-item	58/63	0.00	0.01	0.05	0.10	0.03	0.11	0.07	0.10	0.02	0.00	0.17	0.01	**0.70**
prev-item	53/50	0.00	0.01	0.02	0.04	0.04	0.02	0.08	0.03	0.16	0.00	0.13	0.01	**0.19**
English–French	100/100	0.00	0.34	0.07	0.49	**0.57**	0.47	0.20	0.49	0.13	0.00	0.26	0.00	0.21
English–German	100/100	0.00	0.46	0.14	**0.67**	0.60	0.57	0.43	0.50	0.07	0.00	0.18	0.00	0.02
English–Spanish	100/100	0.00	0.55	0.10	0.76	0.71	0.70	0.38	**0.77**	0.17	0.00	0.38	0.01	0.25
mean		0.01	0.41	0.20	0.55	0.51	0.49	0.45	0.49	0.31	0.16	0.40	0.33	**0.59**
control mean			0.02	0.00	0.02	0.01	0.03	0.02	0.03	0.02	0.03	0.05	0.03	0.04

Table 5: Check 1, task accuracy at the cross-fitted α, mean of the two folds; the main E8 sets and the references (E7 raw +F; E4's CC-20, CC, FV-20, FV-N_t), all cross-fitted over α 1 and 2. Items: patched items in fold 1/2. Control mean: accuracy with the control task's mean. All 31 sets per task: Table 10.

set	check 1	ctrl	check 2	ctrl	heads (median)	tasks
T	0.55	0.02	0.40	0.11	154	14
DB	0.51	0.01	0.32	0.09	65	14
DT	0.49	0.03	0.35	0.12	64	14
DBT	0.49	0.03	0.35	0.14	38	14
BT	0.45	0.02	0.38	0.13	94	14
D	0.41	0.02	0.26	0.05	249	14
TR	0.31	0.01	0.35	0.10	137	14
DTR	0.27	0.04	0.30	0.11	59	14
DBTR	0.25	0.03	0.31	0.12	30	14
DBI	0.25	0.01	0.24	0.11	43	14
DBR	0.22	0.01	0.28	0.08	46	14
BTR	0.22	0.01	0.32	0.10	74	14
DI	0.22	0.01	0.26	0.12	107	14
DTI	0.21	0.03	0.29	0.15	38	14
DR	0.20	0.01	0.25	0.05	229	14
B	0.20	0.00	0.17	0.08	194	14
DBTI	0.18	0.03	0.30	0.15	23	14
TI	0.15	0.01	0.27	0.13	88	14
BTI	0.13	0.01	0.26	0.12	58	14
DTIR	0.13	0.04	0.27	0.15	34	13
R	0.12	0.00	0.06	0.01	958	14
DIR	0.09	0.01	0.23	0.11	98	14
DBTIR	0.09	0.03	0.27	0.14	20	13
DBIR	0.08	0.01	0.22	0.10	35	14
TIR	0.06	0.01	0.24	0.11	80	14
BR	0.06	0.00	0.17	0.07	124	14
I	0.05	0.01	0.06	0.02	329	14
BTIR	0.05	0.01	0.22	0.10	49	14
BI	0.04	0.00	0.13	0.09	120	14
IR	0.03	0.00	0.07	0.02	298	14
BIR	0.00	0.00	0.12	0.09	92	14
E7 raw+F	0.31	0.02	0.34	0.13		
CC-20	0.16	0.03	0.27	0.10		
CC	0.40	0.05	0.34	0.13		
FV-20	0.33	0.03	0.34	0.10		
FV-N_t	0.59	0.04	0.39	0.12		

Table 6: Every set, mean over the 14 tasks, task and control mean in both checks, sorted by check 1; heads: median over tasks; tasks: tasks with a non-empty set.

- **T and DB lead, on different tasks.** T carries the fact tasks (capitals, currency, landmark, park, product) best among the E8 sets and leads DB on German and Spanish; DB carries antonym, synonym, present–past and English–French better. On present–past T has only 13 heads and does nothing; DB carries it, though less than FV-N_t.
- **DB is the best set without a hand-written word list.** With about half CC's heads it is above CC on 9 of 14 tasks and above FV-20 on 9; it stays below FV-N_t on 10 of 14.
- **The BOS filter helps only decoders.** It raises D by 0.10 but lowers T by 0.09; on the other sets it changes little or lowers accuracy. Heads chosen by what they decode need it; heads chosen by where they look do not.
- **T helps most combinations** (12 of 15). Added to DB (DBT, 38 heads), it keeps most of T's fact-task accuracy with a quarter of T's heads, but loses present–past.
- **Input and rest attention hurt.** Adding I or R lowers the mean of every one of the 15 sets it is added to. Alone they do almost nothing.
- **Sequences fail on every E8 set** (next- and prev-item $\leq$0.11); only FV-N_t carries next-item.
- Control means stay low on average. T, BT, DT, DBT and some of their extensions have a high control on park–country (0.18–0.34), so part of their park effect is not task-specific.

set	fold 1, own best α	fold 2, cross-fitted
T	0.55	0.55
DB	0.53	0.49
DT	0.50	0.49
DBT	0.50	0.48
BT	0.46	0.44
D	0.41	0.41
TR	0.30	0.31
DTR	0.29	0.26

Table 7: Filter choice: check-1 accuracy on fold 1 at each task's own best α (top 8 sets), and the same set on fold 2 at the cross-fitted α, mean over tasks.

Chosen on fold 1, T keeps its accuracy on fold 2. DB, DT and DBT lose a little and end level with each other.

5.2 Check 2

Task	items	unp.	D	B	T	DB	DT	BT	DBT	E7 raw+F	CC-20	CC	FV-20	FV-N_t
antonym	100/100	0.00	0.29	0.04	0.37	0.33	**0.39**	0.35	0.36	0.34	0.31	0.35	0.30	0.35
synonym	100/100	0.00	0.11	0.02	0.07	0.16	0.16	0.10	0.14	0.16	0.13	0.17	0.11	**0.18**
singular–plural	11/15	0.00	0.73	0.21	0.76	**0.79**	0.76	0.76	0.75	0.77	0.68	0.77	0.64	0.65
present–past	30/31	0.00	0.68	0.12	0.75	0.69	0.46	0.74	0.46	0.78	0.57	0.64	**0.89**	0.82
country–capital	14/13	0.00	0.52	0.27	**0.81**	0.68	**0.81**	0.76	0.71	0.74	0.73	0.77	0.67	0.74
country–currency	31/34	0.00	**0.37**	0.19	0.29	0.22	0.17	0.25	0.16	0.12	0.12	0.23	0.19	0.17
landmark–country	100/87	0.00	0.04	0.19	0.31	0.22	0.33	0.33	**0.34**	0.19	0.32	0.32	0.33	0.32
park–country	22/23	0.00	0.14	**0.45**	0.38	0.22	0.26	0.32	0.26	0.38	0.21	0.33	0.33	0.38
product–company	84/75	0.00	0.11	0.18	**0.30**	0.16	0.23	0.28	0.22	0.21	0.13	0.16	0.20	0.22
next-item	54/50	0.00	0.02	0.11	0.44	0.12	0.35	0.50	0.55	0.25	0.08	0.34	0.23	**0.61**
prev-item	59/63	0.00	0.03	0.09	0.17	0.05	0.10	0.19	0.10	0.12	0.09	0.09	0.17	**0.32**
English–French	100/100	0.00	0.17	0.15	0.35	0.25	0.32	0.22	0.29	0.22	0.12	0.16	0.26	**0.36**
English–German	100/100	0.00	0.27	0.15	**0.30**	0.27	0.27	0.24	0.24	0.27	0.13	0.21	0.21	0.17
English–Spanish	100/100	0.00	0.21	0.24	**0.35**	0.29	0.33	0.28	0.30	0.25	0.14	0.24	0.20	0.22
mean		0.00	0.26	0.17	**0.40**	0.32	0.35	0.38	0.35	0.34	0.27	0.34	0.34	0.39
control mean			0.05	0.08	0.11	0.09	0.12	0.13	0.14	0.13	0.10	0.13	0.10	0.12

Table 8: Check 2 (1-shot prompts answered wrongly), task accuracy at the cross-fitted α, as Table 5. All 31 sets per task: Table 11.

As in E4 and E7, the sets are close. T is the best E8 set and level with FV-N_t; DB is a little below CC. Sets with T gain most on next-item, but their control means there are close to the task means, so that gain is not task-specific. Several tasks have few items in this check (Table 8), so single-task differences are coarse.

6 The E8 heads and the FV heads

Task	D/FV-20	D/FV-N_t	T/FV-20	T/FV-N_t	DB/FV-20	DB/FV-N_t	DBT/FV-20	DBT/FV-N_t
antonym	13.0 (4.7)	50.0 (27.9)	13.0 (4.0)	59.5 (23.8)	8.5 (1.3)	33.5 (7.7)	7.0 (0.8)	22.0 (5.0)
synonym	11.5 (4.8)	43.5 (28.0)	14.5 (3.1)	49.5 (17.8)	10.0 (1.1)	26.0 (6.4)	9.0 (0.8)	18.0 (4.4)
singular–plural	12.0 (4.5)	48.0 (26.7)	14.0 (2.4)	45.0 (14.3)	12.0 (1.3)	34.0 (7.8)	11.0 (0.6)	20.5 (3.7)
present–past	7.5 (4.3)	31.5 (22.0)	3.0 (0.3)	8.0 (1.3)	6.5 (1.0)	19.0 (5.2)	2.0 (0.1)	3.0 (0.3)
country–capital	14.5 (5.3)	60.5 (34.1)	16.0 (3.9)	58.0 (25.4)	11.5 (1.2)	27.5 (8.1)	10.0 (0.8)	21.0 (5.5)
country–currency	4.5 (5.1)	36.0 (29.5)	8.0 (2.9)	36.5 (16.7)	4.0 (1.0)	17.0 (5.8)	2.5 (0.6)	11.5 (3.6)
landmark–country	16.0 (5.2)	68.0 (33.5)	9.0 (1.7)	29.5 (10.8)	15.0 (1.6)	43.5 (10.7)	7.0 (0.5)	16.0 (3.0)
park–country	15.5 (4.1)	63.5 (30.1)	12.5 (2.2)	37.0 (16.2)	12.5 (1.3)	36.5 (9.7)	8.5 (0.5)	14.5 (3.5)
product–company	14.5 (5.3)	55.5 (33.9)	13.0 (1.1)	28.0 (7.1)	14.0 (1.6)	37.0 (10.2)	10.0 (0.4)	16.5 (2.6)
next-item	9.0 (5.1)	51.0 (32.9)	12.5 (5.9)	71.0 (37.7)	6.5 (1.3)	24.5 (8.0)	6.0 (0.9)	22.0 (5.8)
prev-item	5.5 (4.9)	35.5 (27.1)	9.0 (2.9)	37.0 (16.2)	5.0 (1.4)	23.5 (7.9)	3.0 (0.7)	16.0 (4.1)
English–French	7.0 (5.0)	42.5 (29.6)	6.0 (4.1)	41.5 (24.7)	6.0 (1.3)	23.0 (7.6)	4.5 (0.8)	15.0 (4.8)
English–German	4.5 (4.4)	21.5 (21.5)	9.5 (4.2)	35.0 (20.8)	4.0 (1.1)	13.5 (5.5)	3.0 (0.8)	8.5 (3.8)
English–Spanish	4.5 (4.3)	30.0 (24.3)	7.5 (4.5)	50.5 (25.2)	3.0 (1.2)	15.5 (6.8)	3.0 (0.9)	12.0 (4.9)
mean	10.0 (4.8)	45.5 (28.7)	10.5 (3.1)	41.9 (18.4)	8.5 (1.3)	26.7 (7.7)	6.2 (0.7)	15.5 (3.9)

Table 9: E4's FV-20 and FV-N_t heads (FV-N_t: as many as CC, about 120) found in D, T, DB and DBT, per task, mean of the two folds; in brackets the count expected for a random set of the same size.

DB holds about 6 times as many FV-20 heads as a random set of its size, T about 3 times. DB finds most of FV-20 on the fact tasks except currency, and on singular–plural. On the translations and currency it holds few FV heads, though it carries the translations; there the two selections pick different heads.

7 Takeaways

1. Two cheap selections each come close to the best causally chosen set: **attention to the task words (T)**, and **decoding the task's words while not looking at BOS (DB)**. They carry different tasks.
2. DB uses only the lens and attention, with no hand-written list or causal step, and is above CC and FV-20 on most tasks.
3. Attention to the query or to the rest of the instruction does not pick task heads.
4. How decoded tokens are matched to a word list matters: the leading-space capitalised forms hid most country decoders.

Caveats. One run per setting, greedy, no intervals, no random controls in E8. Set sizes differ (DB about 65 heads, T about 150, FV-20 20, FV-N_t about 120), and earlier work found that size alone moves accuracy; the comparisons are not size-matched. D's list was changed after the first run (Sec. 3). Only α 1 and 2 were tried. T's task-word positions come from a list that includes Yang et al.'s terms. E7's and E4's CC lists still use Yang's expansion and have the same matching gap. Some tasks have few patched items in check 2.

Open next steps.

- Size-matched controls for DB and T: random-drop and layer-matched draws (as E7), and FV sets of exactly DB's size.
- Rebuild E7's raw +F and E4's CC with the case fix, to see how much of their landmark collapse was the gap.
- The union of DB and T, since they win different tasks.
- Why no attention or decoding filter finds the sequence heads that FV-N_t uses on next-item.

Reproduce.

- `scripts/e8_list_length.py`: the list-length check (Table 1).
- `scripts/e8_sets.py -choose A|B`: word lists and sets of each fold.
- `scripts/e8_patch.sh`: patching (GPU, about 45 min on 4 GPUs for all 31 sets); the D rerun with `SETS="D DB ..." PART=d30`.
- `scripts/e8_report.py`: accuracies, `results/e8_filters/report.md`.
- `scripts/e8_tables.py`: these tables.
- First run: commit `4d19133`; its summaries in `results/e8_filters/first_run_d10`.

A All sets, per task

set	ant.	syn.	sg–pl	pr–pa	capital	currency	landm.	park	product	next	prev	En–Fr	En–De	En–Es	mean
D	0.74	0.49	0.80	0.44	0.65	0.28	0.23	0.47	0.21	0.01	0.01	0.34	0.46	0.55	0.41
B	0.00	0.01	0.02	0.02	0.46	0.40	0.77	0.17	0.51	0.05	0.02	0.07	0.14	0.10	0.20
T	0.60	0.18	0.80	0.00	0.98	0.61	0.93	0.67	0.82	0.10	0.04	0.49	0.67	0.76	0.55
I	0.00	0.01	0.00	0.00	0.00	0.00	0.17	0.13	0.28	0.02	0.00	0.09	0.04	0.01	0.05
R	0.00	0.01	0.01	0.01	0.10	0.40	0.42	0.01	0.21	0.02	0.01	0.15	0.07	0.24	0.12
DB	0.70	0.38	0.78	0.54	0.69	0.45	0.73	0.47	0.40	0.03	0.04	0.57	0.60	0.71	0.51
DT	0.65	0.32	0.70	0.00	0.89	0.30	0.89	0.54	0.76	0.11	0.02	0.47	0.57	0.70	0.49
DI	0.37	0.17	0.25	0.36	0.37	0.03	0.17	0.33	0.23	0.00	0.00	0.32	0.27	0.16	0.22
DR	0.48	0.22	0.31	0.07	0.40	0.32	0.02	0.16	0.22	0.01	0.00	0.11	0.21	0.25	0.20
BT	0.34	0.14	0.82	0.00	0.93	0.58	0.88	0.63	0.83	0.07	0.08	0.20	0.43	0.38	0.45
BI	0.00	0.01	0.00	0.00	0.00	0.00	0.12	0.00	0.41	0.01	0.03	0.00	0.00	0.00	0.04
BR	0.01	0.02	0.00	0.00	0.22	0.33	0.13	0.00	0.06	0.03	0.00	0.04	0.01	0.00	0.06
TI	0.00	0.01	0.07	0.00	0.38	0.05	0.34	0.59	0.65	0.04	0.00	0.01	0.01	0.01	0.15
TR	0.22	0.06	0.53	0.00	0.93	0.51	0.76	0.45	0.24	0.05	0.00	0.10	0.12	0.33	0.31
IR	0.00	0.01	0.00	0.00	0.00	0.00	0.03	0.07	0.20	0.03	0.00	0.10	0.01	0.04	0.03
DBT	0.61	0.27	0.80	0.00	0.94	0.27	0.81	0.55	0.74	0.10	0.03	0.49	0.50	0.77	0.49
DBI	0.23	0.23	0.53	0.34	0.30	0.04	0.16	0.31	0.44	0.02	0.01	0.50	0.25	0.10	0.25
DBR	0.30	0.19	0.41	0.01	0.61	0.27	0.16	0.02	0.35	0.02	0.00	0.32	0.13	0.37	0.22
DTI	0.02	0.08	0.38	0.00	0.34	0.04	0.41	0.48	0.61	0.07	0.03	0.30	0.12	0.04	0.21
DTR	0.21	0.11	0.36	0.00	0.69	0.27	0.76	0.28	0.44	0.09	0.00	0.14	0.12	0.34	0.27
DIR	0.22	0.10	0.18	0.03	0.20	0.03	0.05	0.11	0.23	0.00	0.00	0.15	0.01	0.02	0.09
BTI	0.00	0.02	0.06	0.00	0.13	0.01	0.32	0.58	0.72	0.03	0.00	0.00	0.00	0.00	0.13
BTR	0.01	0.03	0.47	0.00	0.81	0.41	0.64	0.30	0.32	0.04	0.00	0.01	0.01	0.03	0.22
BIR	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.01	0.00	0.00	0.00	0.00	0.00
TIR	0.00	0.01	0.02	0.00	0.09	0.03	0.13	0.34	0.21	0.05	0.00	0.00	0.00	0.00	0.06
DBTI	0.02	0.09	0.48	0.00	0.15	0.00	0.29	0.44	0.58	0.07	0.04	0.25	0.04	0.01	0.18
DBTR	0.15	0.09	0.47	0.00	0.73	0.23	0.63	0.20	0.43	0.10	0.00	0.11	0.03	0.30	0.25
DBIR	0.09	0.09	0.33	0.00	0.11	0.01	0.03	0.02	0.29	0.00	0.00	0.10	0.00	0.00	0.08
DTIR	0.01	0.02	0.24	–	0.28	0.04	0.30	0.24	0.39	0.06	0.00	0.04	0.01	0.01	0.13
BTIR	0.00	0.01	0.02	0.00	0.01	0.00	0.02	0.32	0.27	0.01	0.00	0.00	0.00	0.00	0.05
DBTIR	0.00	0.03	0.26	–	0.07	0.00	0.17	0.20	0.36	0.06	0.00	0.02	0.00	0.00	0.09
E7 raw+F	0.46	0.30	0.88	0.17	0.87	0.32	0.03	0.42	0.37	0.02	0.16	0.13	0.07	0.17	0.31
CC-20	0.00	0.09	0.09	0.00	0.77	0.14	0.53	0.40	0.27	0.00	0.00	0.00	0.00	0.00	0.16
CC	0.42	0.34	0.69	0.04	0.75	0.29	0.92	0.73	0.24	0.17	0.13	0.26	0.18	0.38	0.40
FV-20	0.02	0.18	0.44	0.64	0.88	0.00	0.92	0.82	0.64	0.01	0.01	0.00	0.00	0.01	0.33
FV-N_t	0.49	0.55	0.85	0.85	0.94	0.47	0.93	0.88	0.91	0.70	0.19	0.21	0.02	0.25	0.59

Table 10: Check 1, task accuracy at the cross-fitted α, every set and task, mean of the two folds. Blank: empty set.

set	ant.	syn.	sg–pl	pr–pa	capital	currency	landm.	park	product	next	prev	En–Fr	En–De	En–Es	mean
D	0.29	0.11	0.73	0.68	0.52	0.37	0.04	0.14	0.11	0.02	0.03	0.17	0.27	0.21	0.26
B	0.04	0.02	0.21	0.12	0.27	0.19	0.19	0.45	0.18	0.11	0.09	0.15	0.15	0.24	0.17
T	0.37	0.07	0.76	0.75	0.81	0.29	0.31	0.38	0.30	0.44	0.17	0.35	0.30	0.35	0.40
I	0.00	0.01	0.00	0.01	0.00	0.10	0.03	0.48	0.14	0.02	0.05	0.03	0.01	0.03	0.06
R	0.00	0.00	0.00	0.00	0.01	0.25	0.09	0.04	0.13	0.02	0.12	0.05	0.06	0.10	0.06
DB	0.33	0.16	0.79	0.69	0.68	0.22	0.22	0.22	0.16	0.12	0.05	0.25	0.27	0.29	0.32
DT	0.39	0.16	0.76	0.46	0.81	0.17	0.33	0.26	0.23	0.35	0.10	0.32	0.27	0.33	0.35
DI	0.20	0.09	0.71	0.67	0.54	0.26	0.03	0.20	0.12	0.04	0.06	0.23	0.25	0.22	0.26
DR	0.30	0.11	0.73	0.36	0.65	0.36	0.03	0.09	0.13	0.03	0.04	0.16	0.26	0.24	0.25
BT	0.35	0.10	0.76	0.74	0.76	0.25	0.33	0.32	0.28	0.50	0.19	0.22	0.24	0.28	0.38
BI	0.00	0.00	0.02	0.07	0.21	0.22	0.12	0.41	0.19	0.07	0.05	0.12	0.13	0.16	0.13
BR	0.03	0.01	0.22	0.01	0.55	0.17	0.18	0.37	0.19	0.10	0.05	0.14	0.15	0.23	0.17
TI	0.07	0.03	0.53	0.68	0.43	0.24	0.25	0.37	0.30	0.35	0.14	0.14	0.13	0.13	0.27
TR	0.31	0.06	0.76	0.26	0.81	0.34	0.29	0.28	0.30	0.42	0.07	0.36	0.33	0.34	0.35
IR	0.01	0.02	0.00	0.02	0.03	0.06	0.01	0.44	0.12	0.02	0.07	0.05	0.07	0.10	0.07
DBT	0.36	0.14	0.75	0.46	0.71	0.16	0.34	0.26	0.22	0.55	0.10	0.29	0.24	0.30	0.35
DBI	0.24	0.14	0.58	0.51	0.35	0.20	0.17	0.18	0.16	0.08	0.06	0.25	0.25	0.18	0.24
DBR	0.32	0.12	0.74	0.43	0.55	0.23	0.18	0.19	0.17	0.09	0.04	0.25	0.28	0.28	0.28
DTI	0.21	0.13	0.53	0.49	0.67	0.16	0.27	0.23	0.25	0.37	0.07	0.21	0.22	0.21	0.29
DTR	0.36	0.14	0.75	0.11	0.70	0.19	0.33	0.19	0.23	0.30	0.09	0.28	0.24	0.30	0.30
DIR	0.19	0.07	0.68	0.42	0.52	0.24	0.02	0.22	0.12	0.03	0.06	0.21	0.19	0.20	0.23
BTI	0.06	0.04	0.59	0.67	0.36	0.17	0.25	0.29	0.30	0.45	0.13	0.12	0.12	0.12	0.26
BTR	0.23	0.08	0.69	0.26	0.77	0.28	0.34	0.23	0.29	0.48	0.12	0.20	0.26	0.25	0.32
BIR	0.00	0.01	0.00	0.01	0.36	0.16	0.13	0.35	0.17	0.08	0.06	0.12	0.10	0.16	0.12
TIR	0.11	0.04	0.59	0.16	0.48	0.22	0.19	0.31	0.28	0.33	0.07	0.17	0.19	0.17	0.24
DBTI	0.23	0.13	0.56	0.49	0.66	0.14	0.26	0.25	0.23	0.51	0.10	0.21	0.24	0.20	0.30
DBTR	0.34	0.14	0.72	0.11	0.71	0.13	0.33	0.19	0.22	0.53	0.10	0.23	0.26	0.27	0.31
DBIR	0.22	0.10	0.56	0.35	0.39	0.20	0.13	0.18	0.17	0.08	0.05	0.22	0.22	0.17	0.22
DTIR	0.23	0.10	0.50	–	0.69	0.16	0.24	0.18	0.23	0.35	0.11	0.23	0.24	0.21	0.27
BTIR	0.08	0.05	0.60	0.16	0.32	0.14	0.18	0.23	0.29	0.47	0.12	0.16	0.12	0.15	0.22
DBTIR	0.23	0.10	0.52	–	0.69	0.12	0.20	0.20	0.22	0.46	0.12	0.24	0.22	0.18	0.27
E7 raw+F	0.34	0.16	0.77	0.78	0.74	0.12	0.19	0.38	0.21	0.25	0.12	0.22	0.27	0.25	0.34
CC-20	0.31	0.13	0.68	0.57	0.73	0.12	0.32	0.21	0.13	0.08	0.09	0.12	0.13	0.14	0.27
CC	0.35	0.17	0.77	0.64	0.77	0.23	0.32	0.33	0.16	0.34	0.09	0.16	0.21	0.24	0.34
FV-20	0.30	0.11	0.64	0.89	0.67	0.19	0.33	0.33	0.20	0.23	0.17	0.26	0.21	0.20	0.34
FV-N_t	0.35	0.18	0.65	0.82	0.74	0.17	0.32	0.38	0.22	0.61	0.32	0.36	0.17	0.22	0.39

Table 11: Check 2, as Table 10.